

Daily Dose 1 Nov 23 english

DPP: 0

Q1 If $\sec A = \frac{\sqrt{11}}{3}$, then the value of $\frac{\operatorname{cosec}^2 A + \tan^2 A}{\sin^2 A + \cot^2 A}$

is :

- (A) $\frac{13}{9}$
 (B) $\frac{11}{9}$
 (C) $\frac{1}{9}$
 (D) $\frac{11}{7}$

Q2 $99\frac{98}{99} + 99\frac{1}{99}$ is equal to ?

- (A) 200 (B) 199
 (C) $199\frac{1}{99}$ (D) $200\frac{1}{99}$

Q3 In a circle with centre O, PXQ is a tangent at point X. If GH is a diameter such that $\angle QXH = 48^\circ$, then what is the measure of $\angle XHG$?

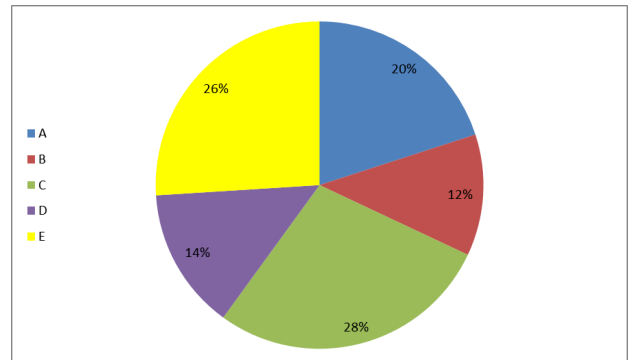
- (A) 32° (B) 42°
 (C) 44° (D) 54°

Q4 Alok lent out a certain sum. He lent $\frac{1}{2}$ part of his sum at 5% SI, $\frac{1}{3}$ part at 6% SI and the remaining part at 8% SI. If Rs. 805 is his total interest after 1 year then find out the money lent out.

- (A) Rs. 13500 (B) Rs. 12600
 (C) Rs. 13800 (D) Rs. 15200

Q5 The following pie-chart represents the distribution of the number of students who appeared for a quiz competition from five different colleges-A, B, C, D, and E in 2021 Read the following data carefully and answer the questions.

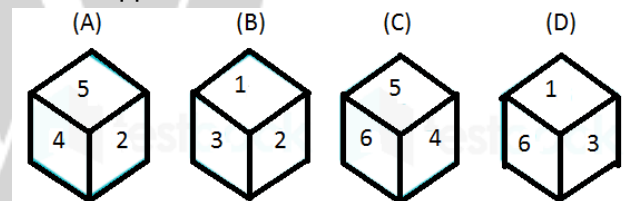
Total number of students who appeared for the quiz = 24000



Find the difference between the number of students who appeared from colleges A and C and the number of students who appeared from colleges B and D:

- (A) 4670 (B) 5280
 (C) 4560 (D) 3450

Q6 Four positions of a dice are given below. What will be the square of the sum of the numbers 3 and its opposite number?



- (A) 25 (B) 36
 (C) 49 (D) 100

Q7 10 cm All the faces of a solid cube of side x are painted black. It is cut into cubical blocks of side 2 cm, then what will be the ratio of the number of cubes whose one surface is black and two surfaces are black?

- (A) 1 : 2 (B) 2 : 3
 (C) 3 : 4 (D) 3 : 2

Q8



Direction: Each of the following questions consists of one statement followed by two arguments. You have to decide which one of them is stronger and which one of them is weak.

Statement:

Should the Ministry of Environment and Forest, on the occasion of World Wildlife day, make an entry to the zoo and sanctuary free for a day.

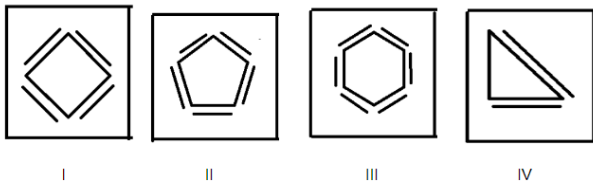
Argument:

I. Yes, because it provides us with an opportunity to propagate our wildlife conservation.

II. No, because it would lead to a huge loss of money from gate receipts.

- (A) Only Argument I is strong.
 (B) Only Argument II is strong.
 (C) Both arguments I and II are strong.
 (D) Neither Argument I nor Argument II is strong.

Q9 Find the odd one out.



- (A) Figure I
 (B) Figure III
 (C) Figure II
 (D) Figure IV

Q10 Q13. Which answer figure will complete the pattern in the following question figure?



(B)



(C)



(D)

Q11 Which of the following statements is NOT true regarding capital expenditure(s)?

- (A) Capital expenditure is transferred to trading and profit and loss accounts.
 (B) Capital expenditures are spread over more than one accounting period.
 (C) Capital expenditure benefits more than one accounting year
 (D) Capital expenditure is incurred to acquire fixed assets for operation of business

Q12 When was world food day Celebrated recently in october 2023?

- (A) 9 October
 (B) 16 October
 (C) 15 October
 (D) 12 October

Q13 A healthy individual has 12-16 grams of haemoglobin in every. *ml* of blood.

- (A) 10000
 (B) 100000
 (C) 100
 (D) 1000

Q14 _____ was the Chairman of the first Finance Commission of India.

- (A) YB Chavan
 (B) AK Chanda
 (C) KC Neogy
 (D) PV Rajamannar

Q15 When was International Rural Women's Day celebrated recently?

- (A) 12 October
 (B) 13 October
 (C) 14 October
 (D) 15 October

Q16



Select the most appropriate option that can substitute the underlined segment in the given sentence. If there is no need to substitute it, select 'No substitution required'.

No sooner he had returned home then he saw the letter lying on the table.

- (A) had he returned home when
- (B) No substitution required
- (C) he returned home than
- (D) had he returned home than

Q17 The following sentence has been split into segments. Identify the segment that contains a grammatical error.

God is great and we could be grateful to him for all the gifts he has bestowed on us.

- (A) No Error
- (B) God is great and
- (C) we could be grateful to him

(D) for all the gifts he has bestowed on us.

Q18 Select the most appropriate SYNONYM of the given word.

Wily

- (A) Frank
- (B) Crafty
- (C) Friendly
- (D) Honest

Q19 Select the option that is closest in meaning to the underlined idiom/phrase in the given sentence.

Please cut down on your expenses now or you will face difficulties in the future.

- (A) recover
- (B) raise
- (C) refuse
- (D) reduce

Q20 Select the option that can be used as a one-word substitute for the given group of words.

A place where antiques are kept

- (A) Museum
- (B) Hangar
- (C) Resort
- (D) Orphanage



Answer Key

Q1 (B)

Q2 (B)

Q3 (B)

Q4 (C)

Q5 (B)

Q6 (C)

Q7 (D)

Q8 (A)

Q9 (D)

Q10 (C)

Q11 (A)

Q12 (B)

Q13 (C)

Q14 (C)

Q15 (D)

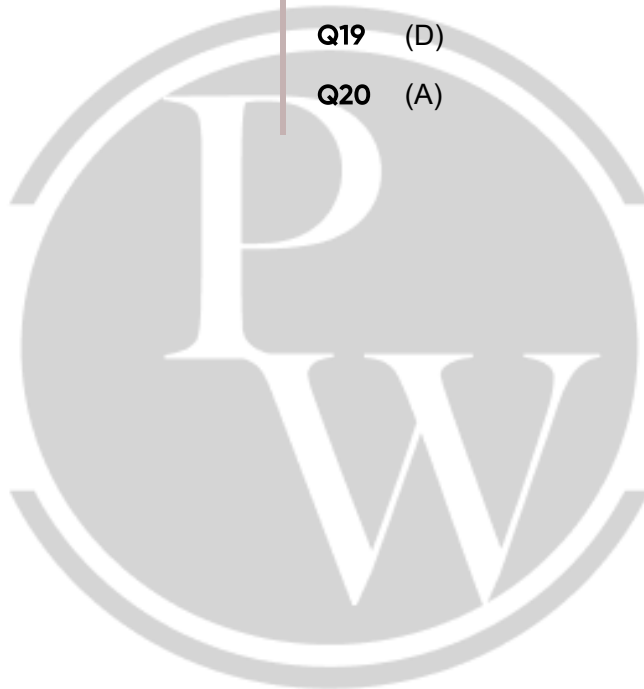
Q16 (D)

Q17 (C)

Q18 (B)

Q19 (D)

Q20 (A)



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Hints & Solutions

Q1 Text Solution:

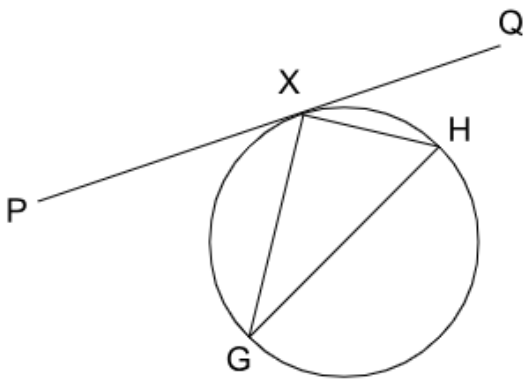
Given, $\sec A = \frac{\sqrt{11}}{3}$

$$\cos = \frac{3}{\sqrt{11}}$$

$$\begin{aligned} \text{Now, } \frac{\operatorname{cosec}^2 A + \tan^2 A}{\sin^2 A + \cot^2 A} &= \frac{\operatorname{cosec}^2 A + \tan^2 A}{\frac{1}{\operatorname{cosec}^2 A} + \frac{1}{\tan^2 A}} \\ &= \operatorname{cosec}^2 A \times \tan^2 A = \frac{1}{\sin^2 A} \times \frac{\sin^2 A}{\cos^2 A} \\ &= \frac{1}{\cos^2 A} = \frac{11}{9} \end{aligned}$$

Q2 Text Solution:

$$\begin{aligned} &99\frac{98}{99} + 99\frac{1}{99} \\ &= 99 + \frac{98}{99} + 99 + \frac{1}{99} \\ &= 99 + 99 + \frac{98}{99} + \frac{1}{99} \\ &= 198 + \frac{98+1}{99} \\ &= 198 + 1 = 199. \end{aligned}$$

Q3 Text Solution:


$$\angle QXH = 48^\circ \text{ (Given)}$$

$$\angle XGH = 48^\circ \text{ (Alternate segment theorem)}$$

$$\angle GXH = 90^\circ \text{ (Angle formed in a semi-circle is a right angle)}$$

In $\triangle GHX$,

$$\angle XGH + \angle XHG + \angle GXH = 180^\circ$$

$$\angle XHG = 180^\circ - (90^\circ + 48^\circ) = 42^\circ$$

Q4 Text Solution:

Let the money lent out to be Rs. P

Now,

$$\begin{aligned} &\frac{\frac{P}{2} \times 5 + \frac{P}{3} \times 6 + \frac{P}{6} \times 8}{\frac{100}{100}} = 805 \\ \Rightarrow &\frac{5P}{2} + 2P + \frac{4P}{3} = 80500 \\ \Rightarrow &\frac{15P + 12P + 8P}{6} = 80500 \\ \Rightarrow &35P = 6 \times 80500 \\ \Rightarrow &P = \frac{6 \times 80500}{35} = 13800 \text{ ₹.} \end{aligned}$$

Q5 Text Solution:

The number of students appeared from colleges A and C

$$= (20 + 28) \% \text{ of } 24000 = 48 = 11520$$

The number of students appeared from colleges B and D

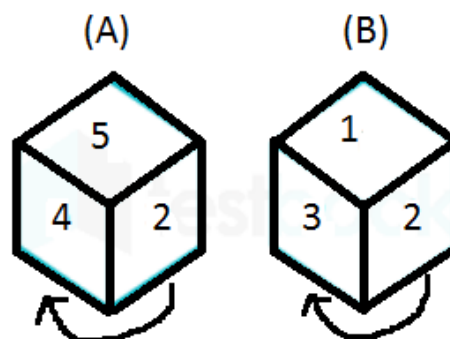
$$= (12 + 14) \% \text{ of } 24000 = 26 = 6240$$

$$\text{Required difference} = 11520 - 6240 = 5280$$

Q6 Text Solution:

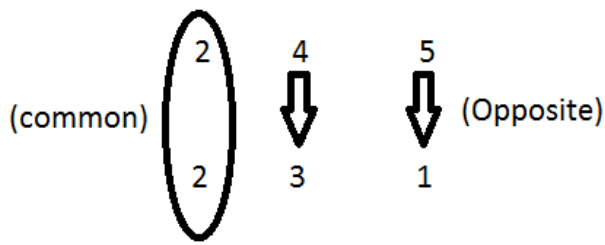
The logic followed here is:

We are taking dice (A) & (B) where 2 is common in both the dice.



Moving clockwise from the common number 5.





Opposite pairs:

$$4 - 3$$

$$5 - 1$$

$$2 - 6$$

So, opposite number of 3 is 4.

$$= 3 + 4 = 7$$

$$= (7)^2 = 49$$

Hence, " 49 " is the correct answer .

Q7 Text Solution:

Each side of the cube n = Side of the big cube / side of the smaller cube

total number of small cube $= n^3$

Number of vertex cubes (cubes with three colored side) = 8

Number of middle cubes (cubes whose two sides are colored) $= 12(n-2)$

Number of central cubes (cubes with one colored sides) $= 6(n-2)^2$

Number of inner central cubes (cubes whose sides are not colored) $= (n-2)^3$

$$n = 10/2$$

$$n = 5$$

One surface is black, such that

$$\begin{aligned} \text{number of cubes} &= 6(n-2)^2 \\ &= 6(5-2)^2 \\ &= 6 \times 3^2 \\ &= 6 \times 9 \\ &= 54 \end{aligned}$$

Two surfaces are black, such that

$$\begin{aligned} \text{number of cubes} &= 12(n-2) \\ &= 12(5-2) \\ &= 12(3) = 36 \end{aligned}$$

According to Question,

The ratio of the number of cubes with one surface black to the number of cubes with two surfaces black

$$54 / 36$$

$$3 : 2$$

Hence , " 3 : 2 " is the correct answer.

Q8 Text Solution:

The intention behind the government's decision to make free entry to the zoo and sanctuary for a day to propagate our ecosystem and wildlife conservation. Hence, Statement I is a strong argument.

Argument II is not strong as the free entry for a day would not really make a huge difference.

Hence , " Only Argument I is strong " is the correct answer .

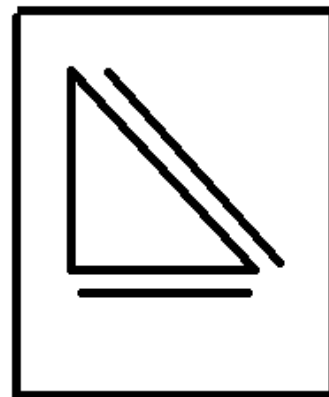
Q9 Text Solution:

In figure I , No. of edges of the figure (4) = No. of lines outside of the figure (4) .

In figure II , No. of edges of the figure (5) = No. of lines outside of the figure (5).

Same for figure III.

But in figure IV, No of edges of the figure (3) $\neq$ No of lines outside of the figure (2).



This is the correct answer.



Hence , "Figure IV" is the correct answer .

Q10 Text Solution:

According to the question :

We have been given a figure which is incomplete .We have to find the correct figure from the options to complete it .

Logic : The figure is divided into 4 equal parts .

The complete figure becomes ,



Hence , " " is the correct answer .

Q11 Text Solution:

- Capital expenditure refers to the expenses incurred by a business to acquire or improve fixed assets such as land, buildings, equipment, machinery, vehicles, etc., which are used for the operation of the business.
- These assets provide long-term benefits to the business and are not intended for sale in the near future.
- Capital expenditure is recorded on the balance sheet and is not immediately recognized as an expense in the income statement.

- Capital expenditure is not transferred to the trading and profit and loss account as it is not recognized as an expense in the income statement.
- Instead, it is depreciated over the useful life of the asset and the depreciation expense is recognized in the income statement as an expense.
- Capital expenditures are spread over more than one accounting period as the benefits from these expenditures are enjoyed over several years.
- These expenditures are therefore capitalized and depreciated over their useful lives.

Hence, the correct answer is "Capital expenditure is transferred to trading and profit and loss account".

Q12 Text Solution:

- World Food Day 2023 is observed annually on the 16th of October 2023.
- The theme of World Food Day is - Water is life, water is food. Leave no one behind.
- The Food and Agriculture Organization (FAO), a specialized body of the United Nations, created World Food Day in 1945.
- However, it would take another 34 years until the 20th FAO conference in November 1979, when it was declared a global holiday.
- Following this, 150 nations continued to commemorate the day once it was given official UN recognition.
- Since 2014, the day's popularity has been exploited to advance the causes of world hunger relief and eradicate poverty in developing countries.

Hence the correct answer is 16 October

Q13 Text Solution:



- A healthy individual has 12-16 grams of hemoglobin in every 100 ml of blood.
- Hemoglobin is a globular protein present in RBC which transports oxygen through blood in our body.
- From the lungs to various parts of the body, Hemoglobin helps in transporting O_2 as oxy-hemoglobin and it is a respiratory pigment.
- CO_2 is also transported back as carbamino hemoglobin via Hemoglobin.
- Hemoglobin levels per dl of blood are following:-

Age	Level of Hemoglobin(g/dl)
New born	17-22
Children	11-13
Adult male	14-18
Old male	12.4-14.9

Hence the correct answer is 100.

Q14 Text Solution:

- The first Finance Commission of India was constituted in 1951 and its chairman was KC Neogy.
- The commission was set up to make recommendations on how to allocate financial resources between the center and the states.
- Neogy was a senior Indian Civil Service officer who had served in various capacities in the government of India, including as the financial adviser to the governor of Bihar.
- He was known for his expertise in public finance and played a key role in the drafting of the Constitution of India.
- During his tenure as the chairman of the first Finance Commission, Neogy and his team visited various parts of the country to

understand the financial needs and challenges of different states.

- The recommendations made by the commission had a significant impact on the fiscal relations between the center and the states in India.
- From 1979 to 1980, YB Chavan, an Indian politician, held the position of Deputy Prime Minister.
- AK Chanda was an Indian civil servant who served as the first Chief Election Commissioner of India from 1950 to 1958. He was also a member of the Constituent Assembly of India.
- PV Rajamannar was an Indian civil servant and served as the Governor of the Reserve Bank of India from 1949 to 1957.

Hence, the correct answer is "KC Neogy".

Q15 Text Solution:

- The U.N.'s efforts to improve the status and well-being of women, particularly those in rural areas, rank among its most impressive accomplishments
- It was proposed to empower and recognize women in 1995 during the Fourth World Conference on Women in Beijing, China.
- In order to recognize the importance of rural women to agriculture, food production, and food safety, it was suggested that October 15 be observed as International Day of Rural Women as October 16 was World Food Day
- A resolution calling for the development of ICT skills for all, especially women and indigenous people living in remote and rural communities, was adopted in 2003 at the World Summit on the Information Society in Geneva.

Hence the correct answer is 15 October



Q16 Text Solution:

The underlined part of the sentence is grammatically correct and is related to "inversion".

- "No sooner....than" is the expression used in the sentence which is a negative adverb.
- As per the rule of inversion when using negative adverbs at the start of the sentence we need to use the auxiliary verb just after the adverb.
 - e.g. No sooner had she finished her performance than the lights went out.

Correct sentence: No sooner had he returned home than he saw the letter lying on the table.

Hence, the correct answer is "had he returned home than".

Q17 Text Solution:

The given sentence is grammatically incorrect and the error lies in the option "we could be grateful to him". The error is related to the "modal verbs".

- The modal verb "**could**" implies a possibility or a conditional situation, but the sentence is not expressing either of those meanings.
- The given sentence states a fact or an obligation: we should be grateful to God for all the gifts he has given us.
- Therefore, the modal verb "**should**" would be more appropriate in this context.
 - As it is used to express obligations or duties politely.

Hence, we should replace "and we could" with "and we should" to make the sentence grammatically correct.

Correct sentence: God is great and we should be grateful to him for all the gifts he has bestowed on us.

Q18 Text Solution:

Wily: skilled at gaining an advantage, especially deceitfully.

Let us first understand the given options :

- **Frank**: open, honest, and direct in speech or writing, especially when dealing with unpalatable matters.
- **Crafty**: clever at achieving one's aims by indirect or deceitful methods.
- **Friendly**: kind and pleasant.
- **Honest**: free of deceit and untruthfulness; sincere.

Hence, "Crafty" is the most appropriate synonym for the word "Wily".

Q19 Text Solution:

The idiom "cut down on" means to reduce the amount or consumption of something, typically in the context of food, expenses, or other resources.

- E.g.:To improve his health, he decided to cut down on his sugar intake and started consuming fewer sugary snacks and drinks.

Hence, the correct answer is "reduce".

Q20 Text Solution:

Let us first understand the given options :

- **Museum**: a building in which objects of historical, scientific, artistic, or cultural interest are stored and exhibited.
- **Hangar**: a large building with an extensive floor area, typically for housing aircraft.
- **Resort**: a place that is a popular destination for vacations or recreation, or which is frequented for a particular purpose.
- **Orphanage**: a residential institution for the care and education of orphans.

Hence, the correct answer is "museum".

